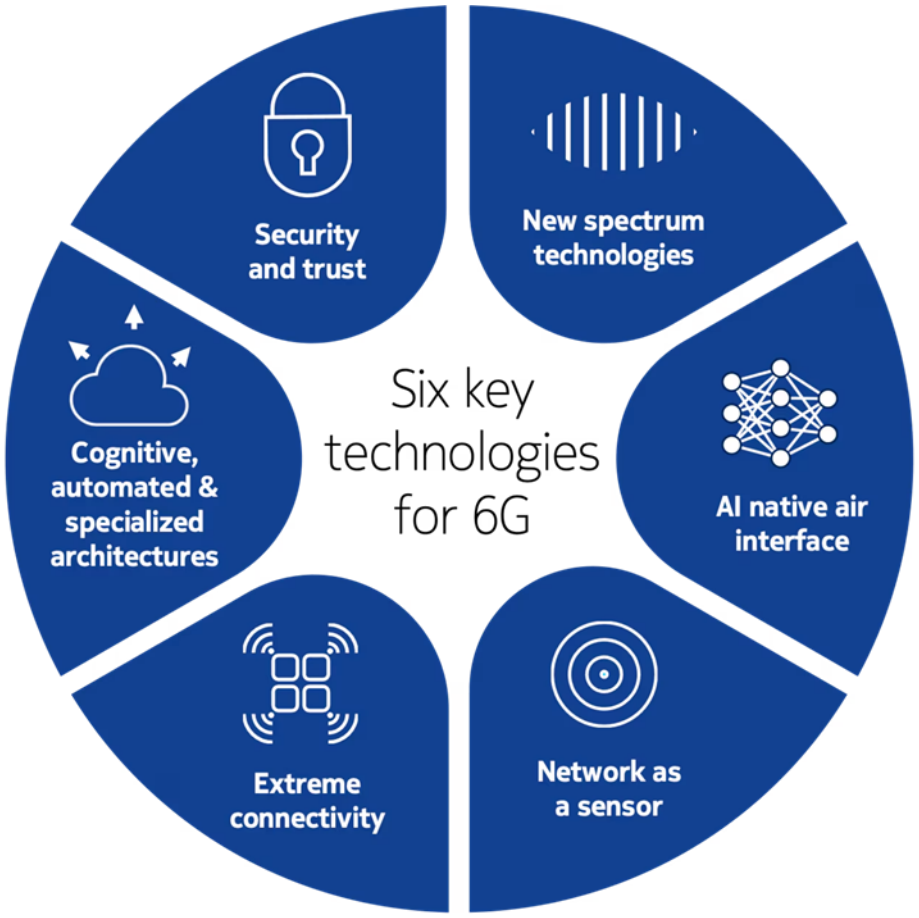
Optimizing Edge-AI for 6G via Split and Federated Learning Paradigms

MECT Thesis

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- *Supervisor: Arnaldo Oliveira*
- *Co-supervisor: Fábio Coutinho*

universidade de aveiro

Context



Source: [1]

6G networks need to support

Holographic comms.	Low-latency Industrial IoT	Ubiquitous connectivity	Autonomous vehicles

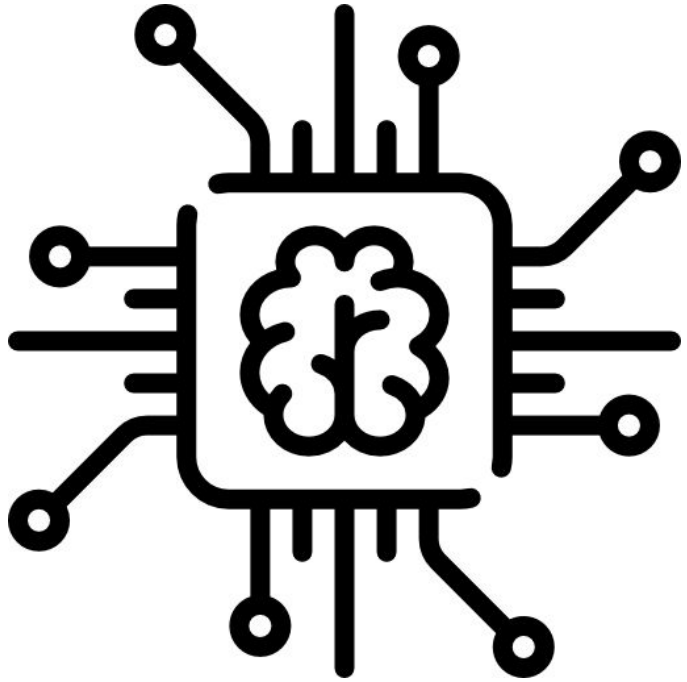
This means 6G networks must **cut latency**, **boost bandwidth** and **protect data** better than 5G

AI will be **crucial**: networks must be **dynamic** & **intelligent**

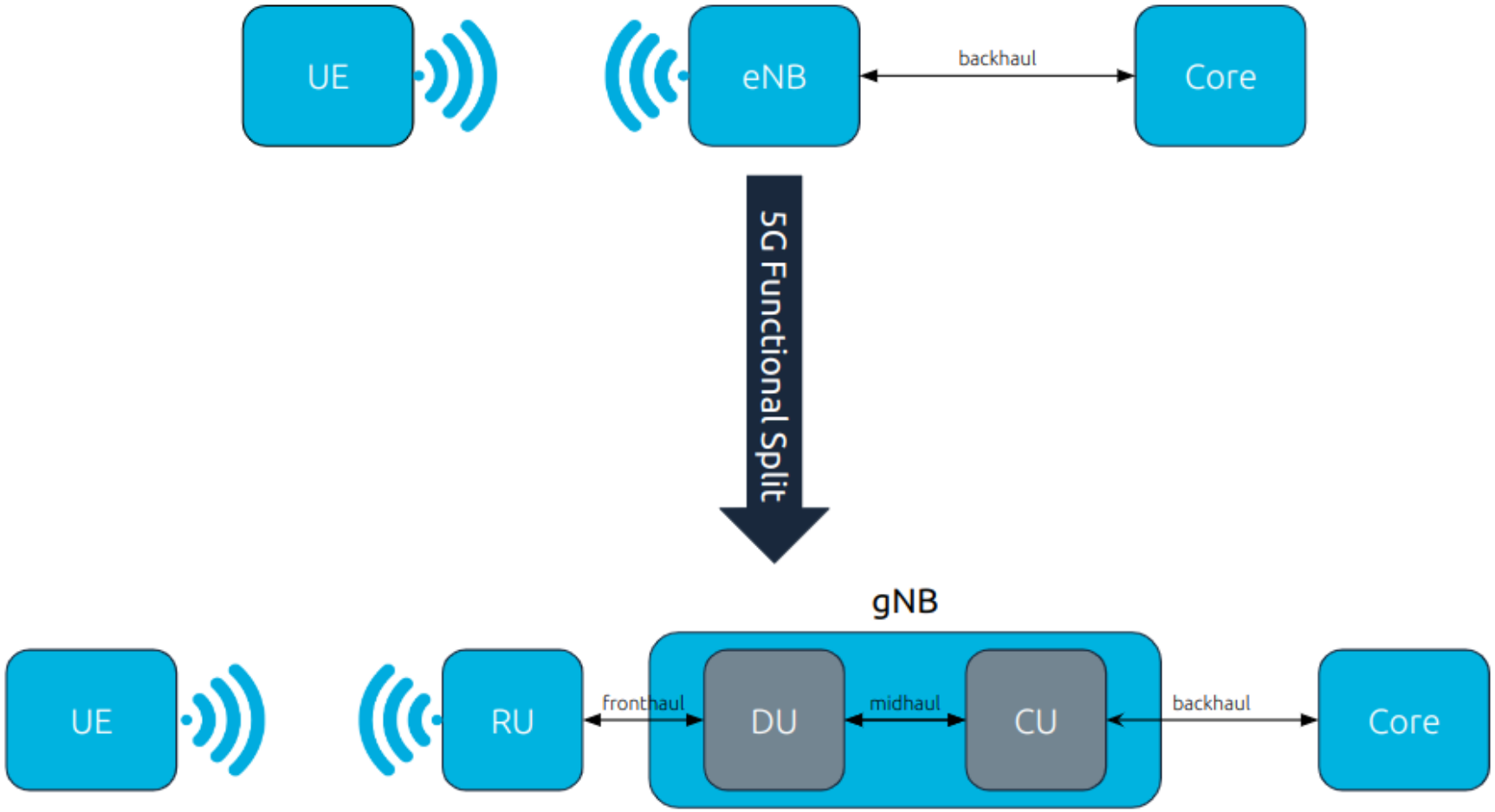
Problem and Motivation

Traditional **centralized AI won’t scale** with 6G **data & devices**

Intelligence needs to **be distributed** throughout the network



ORAN Architecture Overview

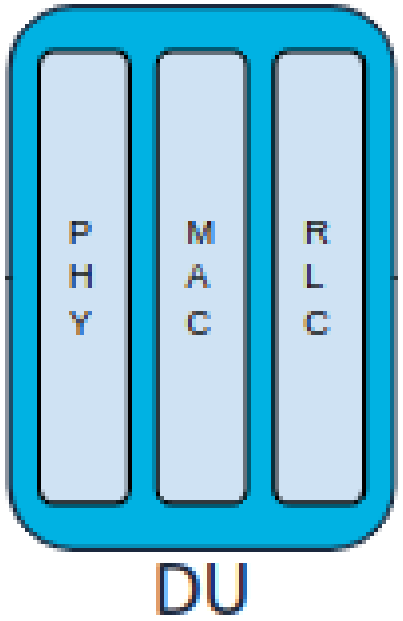


Source: [2]

ORAN architecture overview

DU functions

- **PHY**
  - **Radio signal transmission and reception**
- **MAC**
  - **User scheduling and radio resource management**
- **RLC**
  - **Data segmentation, reassembly and retransmission**

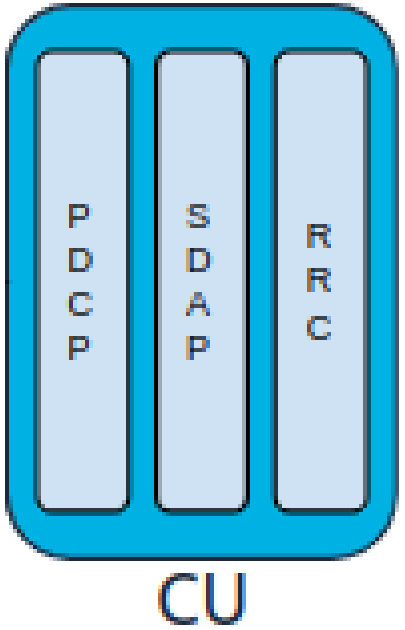


Source: [3]

ORAN architecture overview

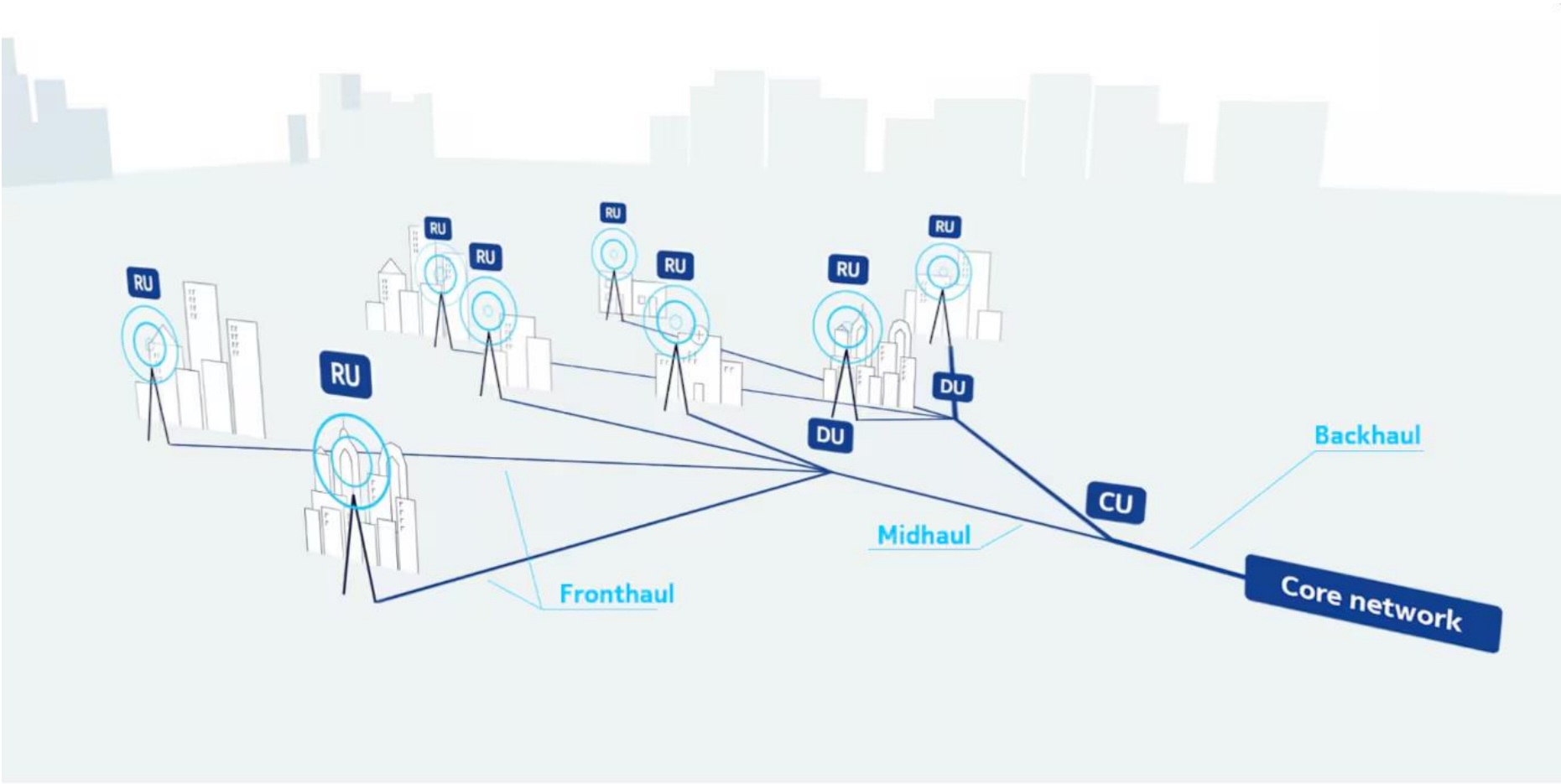
CU functions

- **PDCP**
  - **Encryption and integrity protection**
- **SDAP**
  - **QoS flow to radio bearer mapping**
- **RRC**
  - **Connection, mobility control and radio resource configuration**



Source: [3]

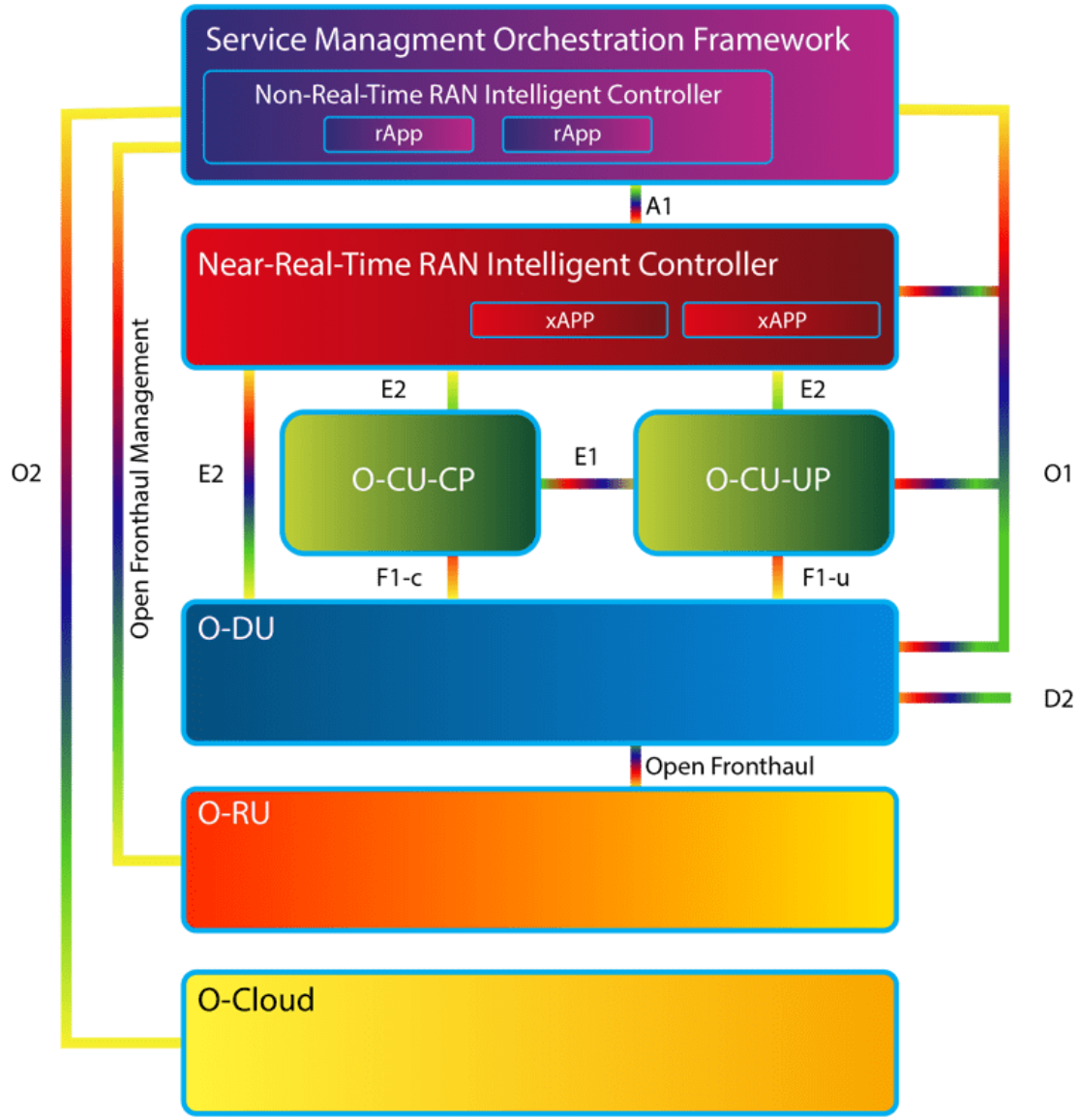
ORAN Architecture Overview



Source: [4]

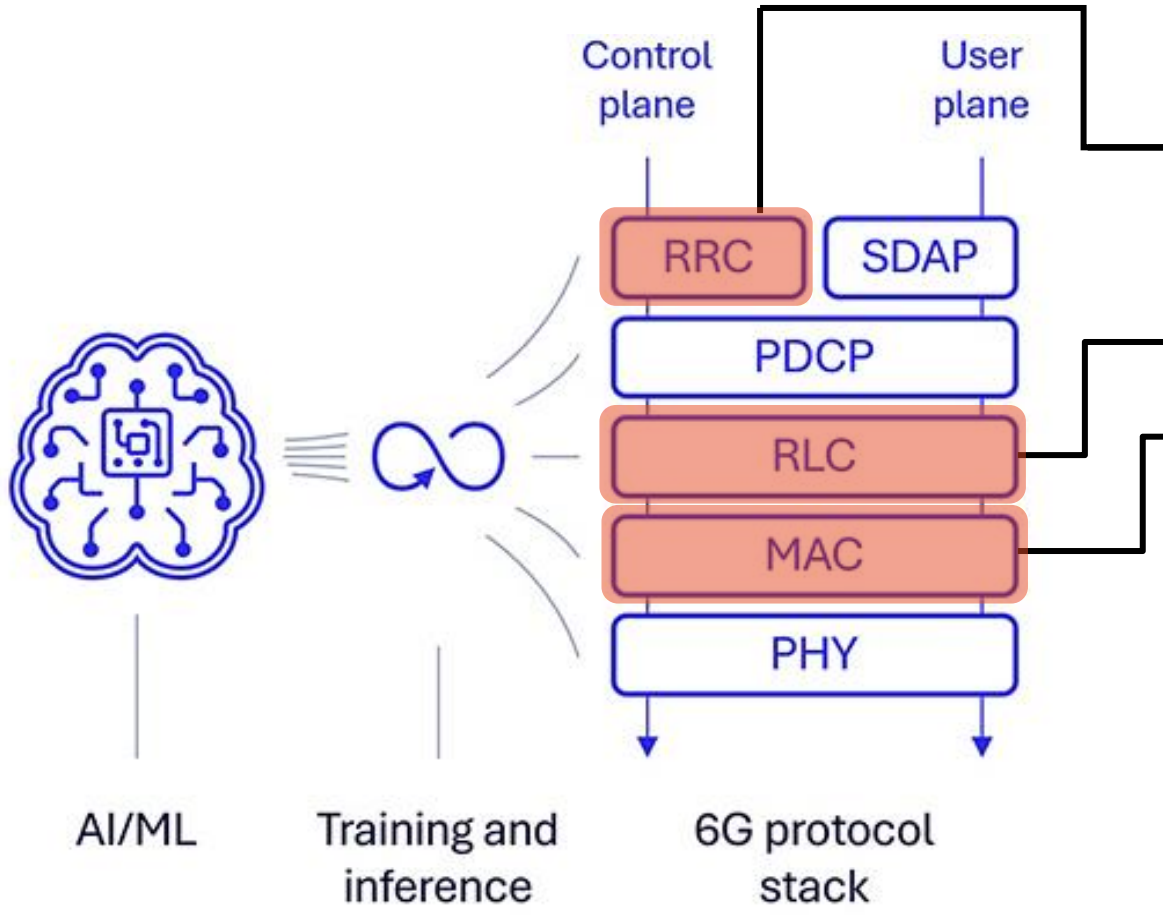
O-RAN alliance interface definition

- xApps
  - Used for optimizing radio resources in real time
- rApps
  - Used for long-term optimization, policy management, data analytics, and ML model training



Source: [5]

Possible Network Optimizations



- Network conditions prediction (Radio Resource Control)
- Resource allocation optimization
- Error Detection (Radio Link Control)
- Shared medium access management (Medium Access Control)

➢ **AI can proactively improve network performance across multiple protocol layers**

Source: [6]

Conclusion



Source: [7]

Thesis goal

- **Deploy** and **evaluate** multiple **network architectures**, leveraging **ML algorithms** for **performance optimization**

Key Outcomes

- Demonstrate how **AI improves network performance**
- Show improvements in **latency, bandwith and privacy**
- Provide **pratical use cases** where edge intelligence adds value

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Questions?